

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Examination, 2013-14

Paper No. MTB-201 (Old) / MTB-202 (New): Statics and Dynamics

Time: 3 Hours

Full Marks: 70

Code: BS/Sem II/3

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory.)

- (a) Forces with components $(-150, 0)$, $(30, 40)$, $(0, -90)$, $(120, 50)$ act at $(-6, 1)$, $(-3, 6)$, $(5, 0)$, $(0, -5)$ respectively. Show they are in equilibrium. [2]
- (b) Three forces P, Q, R act along sides BC, CA, AB of $\triangle ABC$. If their resultant passes through the circumcentre, show $P \cos A + Q \cos B + R \cos C = 0$. [2]
- (c) If the distance between two particles remains unchanged during a displacement, find the sum of works done by their mutual action and reaction. [2]
- (d) If the angular velocity of a moving point about a fixed origin is constant, show its transverse acceleration varies as its radial velocity. [2]
- (e) Radial and transverse velocity components of a particle are λr and $\mu \theta$. Find the path. [2]
- (f) If the path is a circle with the centre of force at the centre, find the law of force. [2]
- (g) State Kepler's second law. Show the transverse component of a planet's acceleration is zero. [2]
- (h) In SHM, show the average acceleration is obtained by multiplying its maximum by $\frac{2}{\pi}$. [2]
- (i) If a particle describes a circle with constant angular velocity, prove the foot of the perpendicular from the particle on any diameter executes SHM. [3]
- (j) A particle moves on a smooth curve under gravity with velocity proportional to the arc distance from the highest point. Prove the curve is a cycloid. [3]

Question 2.

- (a) Show a system of coplanar forces can be reduced to a single force through a given point and a couple. [6]
- (b) Forces in the plane of $\triangle ABC$ are equivalent to a force at A along the bisector of $\angle BAC$ and couple G_1 . Moments about B, C are G_2, G_3 . Show $(b + c)G_1 = bG_2 + cG_3$. [6]

Question 3.

- (a) State and prove the principle of virtual work for a system of coplanar forces. [6]
- (b) A string of length a forms the shorter diagonal of a rhombus of four uniform rods each of length b and weight W , hinged together, one rod horizontal. Prove the tension in the string is $\frac{2W(2b^2 - a^2)}{b\sqrt{4b^2 - a^2}}$. [6]
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Question 4.

- (a) Obtain the tangential and normal components of acceleration of a particle moving in a plane curve. [6]
- (b) If radial and transverse velocities are always proportional, and the same holds for accelerations, prove the velocity varies as some power of the radius vector. [6]
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Question 5.

- (a) A heavy bead slides on a smooth circular wire of radius a , projected from the lowest point with velocity just sufficient to reach the highest point. Show the radius through the bead turns through angle $2 \tan^{-1} \left(\sinh \left(t \sqrt{g/a} \right) \right)$ in time t . [6]
- (b) A particle slides down a smooth cycloid from rest at the cusp, axis vertical, vertex down. Show at the vertex the pressure on the curve equals twice the weight. [6]
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Question 6.

- (a) If the central orbit is $r^n = a^n \cos n\theta$ under force toward the pole, find the law of force. [6]
- (b) A particle moves under central acceleration $\mu/(\text{distance})^2$, projected with velocity V at distance R . Show its path is a rectangular hyperbola if angle of projection is $\sin^{-1} \frac{\mu}{V \sqrt{V^2 R^2 - 2\mu R}}$. [6]
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Question 7.

- (a) Discuss the motion of a particle in a frame moving in a straight line with acceleration relative to a fixed Newtonian frame. [6]
- (b) A particle performs SHM of period T about centre O . It passes through P (where $OP = b$) with velocity v in the direction of OP . Show the time to return to P is $\frac{T}{\pi} \tan^{-1} \frac{2\pi bv}{v^2 T^2 - 4\pi^2 b^2}$. [6]
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